MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 03** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

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Q1. A ladder makes an angle of 60° with the ground when placed against a wall. If the foot of the ladder is 2 m away from the wall, then the length of the ladder (in metres) is: [1]

(A) $\dfrac{4}{\sqrt{3}}$ (B) $4\sqrt{3}$ (C) $2\sqrt{2}$ (D) 4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (D)

Justification: $\cos 60^\circ = \dfrac{2}{L} \implies \dfrac{1}{2} = \dfrac{2}{L} \implies L = \boxed{4}$.

Q2. If the angle of elevation of a tower from a distance of 100 m from its foot is 60°, then the height of the tower is: [1]

(A) $100\sqrt{3}$ m (B) $\dfrac{100}{\sqrt{3}}$ m (C) $50\sqrt{3}$ m (D) $\dfrac{200}{\sqrt{3}}$ m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\tan 60^\circ = \dfrac{h}{100} \implies h = 100 \tan 60^\circ = \boxed{100\sqrt{3} \text{ m}}$.

Q3. The tops of two poles of heights 16 m and 10 m are connected by a wire of length l metres. If the wire makes an angle of 30° with the horizontal, then l is: [1]

(A) 26 m (B) 16 m (C) 12 m (D) 10 m

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: Vertical height difference $= 16 - 10 = 6$ m. Thus $\sin 30^\circ = \dfrac{6}{l} \implies \dfrac{1}{2} = \dfrac{6}{l} \implies l = \boxed{12 \text{ m}}$.

Q4. If a 1.5 m tall girl stands at a distance of 3 m from a lamp-post and casts a shadow of length 4.5 m on the ground, then the height of the lamp-post is: [1]

(A) 1.5 m (B) 2 m (C) 2.5 m (D) 2.8 m

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C)

Justification: By similar right triangles, $\frac{H}{3+4.5} = \frac{1.5}{4.5} \implies \frac{H}{7.5} = \frac{1}{3} \implies H = \boxed{2.5 \text{ m}}$.

Q5. **Assertion (A):** If the angles of elevation of the top of a tower from two points at distances of 9 m and 16 m from the base in the same straight line are 30° and 60°, then the height of the tower is 12 m. **Reason (R):** When the angles of elevation of the top of a tower from two points on the same side at distances a and b are complementary, the height of the tower is given by $h = \sqrt{ab}$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Here $30° + 60° = 90°$ (complementary) and $h = \sqrt{9 \times 16} = \sqrt{144} = 12$ m. Both (A) and (R) are true and (R) correctly explains (A).

SECTION B: Short Answer Questions

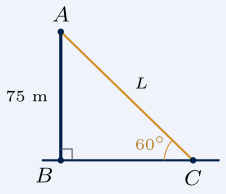
MATHS MANTHAN | (11 Marks)

Q6. A kite is flying at a height of 75 m above the ground, attached to a string inclined at 60° to the horizontal. Find the length of the string, assuming there is no slack. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let kite be at A at height $AB = 75$ m above ground point B, tied to string AC inclined at $\angle ACB = 60°$. [0.5 Mark]

In right $\triangle ABC$: $\sin 60° = \frac{AB}{AC} \implies \frac{\sqrt{3}}{2} = \frac{75}{AC}$. [1 Mark]



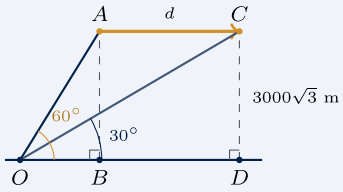
$$AC = \frac{150}{\sqrt{3}} = 50\sqrt{3} = 50(1.732) = \boxed{50\sqrt{3} \text{ m} \approx 86.60 \text{ m}}$$

Final string length: [0.5 Mark]

Q7. The angle of elevation of an aeroplane from a point on the ground is 60°. After a flight of 30 seconds, the angle of elevation becomes 30°. If the aeroplane is flying at a constant height of $3000\sqrt{3}$ m, find the speed of the aeroplane in km/h. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let aeroplane move horizontally from A to C at height $AB = CD = 3000\sqrt{3}$ m, viewed from O with $\angle AOB = 60°$ and $\angle COD = 30°$. [0.5 Mark]



In right $\triangle ABO$: $\tan 60° = \dfrac{3000\sqrt{3}}{OB} \implies OB = 3000$ m. In right $\triangle CDO$: $\tan 30° = \dfrac{3000\sqrt{3}}{OD} \implies OD = 3000\sqrt{3} \cdot \sqrt{3} = 9000$ m. [1 Mark]

Distance covered in 30 s: $AC = BD = OD - OB = 9000 - 3000 = 6000$ m. [1 Mark]

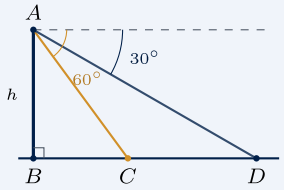
$$\text{Speed} = \frac{6000 \text{ m}}{30 \text{ s}} = 200 \text{ m/s} = 200 \times \frac{18}{5} \text{ km/h} = \boxed{720 \text{ km/h}}$$

Converting speed to km/h: [0.5 Mark]

Q8. A man on a cliff observes a boat at an angle of depression of 30° approaching the shore immediately beneath the observer with uniform speed. Six minutes later, the angle of depression of the boat is found to be 60°. Find the total time taken by the boat to reach the shore. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let cliff $AB = h$ and boat sail from D ($\angle ADB = 30°$) to C ($\angle ACB = 60°$) towards shore B at uniform speed v. [0.5 Mark]



In right $\triangle ABC$: $\tan 60° = \dfrac{h}{BC} \implies BC = \dfrac{h}{\sqrt{3}}$. In right $\triangle ABD$: $\tan 30° = \dfrac{h}{BD} \implies BD = h\sqrt{3}$. [1 Mark]

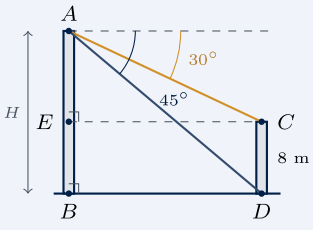
Distance $DC = BD - BC = h\sqrt{3} - \dfrac{h}{\sqrt{3}} = \dfrac{2h}{\sqrt{3}}$ is covered in 6 min. Thus uniform speed $v = \dfrac{2h}{6\sqrt{3}} = \dfrac{h}{3\sqrt{3}}$ units/min. [1 Mark]

Remaining time from C to $B = \dfrac{BC}{v} = \dfrac{h/\sqrt{3}}{h/(3\sqrt{3})} = 3$ minutes. Therefore, total time from start point D to shore $B = 6 + 3 = \boxed{9 \text{ minutes}}$. [0.5 Mark]

Q9. The angles of depression of the top and bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45°, respectively. Find the height of the multi-storeyed building and the distance between the two buildings. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let multi-storeyed building $AB = H$ m and 8 m building CD. Draw $CE \perp AB$ so $BE = CD = 8$ m and $AE = H - 8$. By alternate angles, $\angle ACE = 30°$ and $\angle ADB = 45°$. [0.5 Mark]



In right $\triangle ABD$: $\tan 45° = \dfrac{H}{BD} \implies BD = AE_{\text{horiz}} = CE = H$ — (i). In right $\triangle AEC$: $\tan 30° = \dfrac{H-8}{CE} \implies CE = \sqrt{3}(H-8)$ — (ii). [1 Mark]

Equating (i) and (ii): $H = \sqrt{3}(H-8) \implies H(\sqrt{3}-1) = 8\sqrt{3} \implies H = \dfrac{8\sqrt{3}(\sqrt{3}+1)}{2} = 4(3+\sqrt{3})$ m. [1 Mark]

Since distance between buildings $BD = H$, substituting $\sqrt{3} = 1.732$ gives $4(4.732) = 18.928$ m.

$$\therefore \text{ Height} = \text{Distance} = \boxed{4(3+\sqrt{3}) \text{ m} \approx 18.928 \text{ m}}$$

Both required values: [0.5 Mark]

Q10. CASE STUDY: Telecommunication Transmission Tower Engineering

A radio transmission tower has two vertical sections on level ground: lower transmission mast Section B (foot P, top B) and upper microwave relay Section A mounted directly on top of B (top A). Observation station O on the ground is at $OP = 36$ m from base P, with elevation of top of B being $30°$ and top of A being $45°$.

(i) Find the height of the lower transmission mast Section B (i.e., height PB). [1]

(ii) Find the total height of the combined tower from ground to the top of Section A (i.e., height PA). [1]

(iii) Calculate the height of Section A alone (i.e., vertical distance AB). [Take $\sqrt{3} = 1.732$] [2]

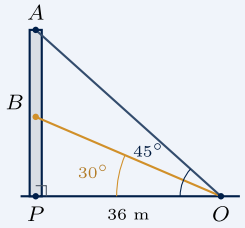
OR

Find the length of the supporting cable running from observation station O to the top of Section B. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) In right $\triangle OPB$ ($\angle BOP = 30°, OP = 36$ m):

$$\tan 30° = \frac{PB}{36} \implies PB = \frac{36}{\sqrt{3}} = \boxed{12\sqrt{3} \text{ m} \approx 20.784 \text{ m}}$$



Subpart (i): [1 Mark]

(ii) In right $\triangle OPA$ ($\angle AOP = 45°$):

$$\tan 45° = \frac{PA}{36} \implies 1 = \frac{PA}{36} \implies PA = \boxed{36 \text{ m}}$$

Subpart (ii): [1 Mark]

(iii) Main Choice: Height of Section A alone is $AB = PA - PB = 36 - 12\sqrt{3}$ m. [1 Mark]

Using $\sqrt{3} = 1.732$: $AB = 36 - 12(1.732) = 36 - 20.784 = \boxed{15.216 \text{ m}}$. [1 Mark]

(iii) OR Choice: In right $\triangle OPB$, supporting cable length OB satisfies $\cos 30° = \frac{OP}{OB}$. [1 Mark]

$$\implies OB = \frac{36}{\sqrt{3}/2} = \frac{72}{\sqrt{3}} = \boxed{24\sqrt{3} \text{ m} \approx 41.57 \text{ m}}$$

Cable length OB: [1 Mark]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★